

Robust Transmit Beampattern Design of Colocated MIMO Radar for Multi-Target Tracking under Parameter Estimation Uncertainty

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Abstract—Radar resource scheduling in multi-target tracking (MTT) tasks typically relies on estimated parameters from the previous frame as prior information to guide current resource allocation. However, the uncertainty in angle estimation leads to beam pointing misalignment, resulting in severe degradation of transmit gain. To mitigate target angular estimation uncertainty, this paper models it using the prediction covariance of the extended Kalman filter (EKF) and derives the robust Bayesian Cramér-Rao lower bound (RBCRLB) as a performance metric. Based on this metric, a robust transmit beampattern design (RTBD) problem is formulated and transformed into a semidefinite programming (SDP) problem via angular discretization for efficient solving. Simulation results verify that the proposed strategy effectively improves tracking performance in the presence of angular estimation errors.

Index Terms—Colocated MIMO radar, resource scheduling, multi-target tracking, robust transmit beampattern design.

I. INTRODUCTION

Colocated multiple-input multiple-output (C-MIMO) radar systems have attracted significant attention due to the transmit degrees of freedom provided by waveform diversity. By optimizing the waveform correlation matrix, C-MIMO radar can flexibly synthesize transmit beampatterns [1], [2], demonstrating superior situational awareness potential in multi-target tracking (MTT) tasks compared to traditional phased-array radars. However, given the limited system resources such as transmit power and time aperture, efficient resource scheduling is essential to maintain stable tracking performance in dynamic environments [3]. Classical resource scheduling algorithms typically rely on ideal a priori assumptions, treating the predicted target positions as true values, and optimize multidimensional resources such as power, beam allocation, and dwell time [4]–[8] to maximize tracking accuracy.

In practical scenarios, non-ideal factors such as noise and clutter inevitably degrade observation quality. When processed

through tracking filters, these corrupted measurements, combined with inherent process noise in MTT, introduce errors into the predicted state parameters [9], [10]. Addressing this, [9] investigated how parameter estimation uncertainty degrades tracking performance bounds via the Jacobian matrix. However, this research primarily focuses on theoretical performance limits and overlooks the physical beam pointing misalignment caused by prediction errors, which significantly reduces the transmit gain at the true target angles. To ensure beam coverage, [11] adopted a beam scanning strategy, which essentially sacrifices time resources for spatial coverage. Furthermore, although studies such as [12]–[15] considered signal-level beampattern synthesis for MTT, they assumed perfectly known target angles. By failing to account for the impact of angular estimation errors on beam pointing, these methods exhibit a lack of robustness in practical applications.

Motivated by these limitations, this paper proposes an uncertainty-aware robust transmit beampattern design (RTBD) strategy for MTT. Unlike general robust resource allocation frameworks, the fundamental contribution of this work lies in directly incorporating EKF-based angular uncertainty into the physical spatial beam synthesis. Specifically, using the EKF prediction covariance to bound the angular uncertainty, we derive the robust Bayesian Cramér-Rao lower bound (RBCRLB) as a tractable performance metric. By discretizing the angular intervals, an SDP problem is formulated to minimize the worst-case RBCRLB. The synthesized beampattern effectively covers the uncertainty intervals, directly preventing physical beam pointing misalignment and enhancing tracking stability.

II. SYSTEM MODEL

A. Target Motion Model

Consider Q point targets performing nearly constant velocity (CV) motion in a 2-D Cartesian plane. The state vector $\mathbf{x}_{q,k}$ of target q at time k evolves as [10]:

$$\mathbf{x}_{q,k} = \mathbf{F}\mathbf{x}_{q,k-1} + \mathbf{u}_{q,k-1} \quad (1)$$

This work was supported in part by the National Natural Science Foundation of China under Grant 62301127, in part by the Natural Science Foundation of Sichuan Province under Grant 2026NSFSC1422, in part by the Fundamental Research Funds for the Central Universities under Grant ZYGX2025XJ004.

where $\mathbf{F}$ is the state transition matrix. The process noise vector $\mathbf{u}_{q,k-1}$ accounts for the uncertainty in target motion and is assumed to follow a zero-mean Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{Q}_{q,k-1})$. The process noise covariance matrix is expressed as:

$$\mathbf{Q}_{q,k-1} = \kappa_q \mathbf{I}_2 \otimes \begin{bmatrix} \frac{T^3}{2} & \frac{T^2}{2} \\ \frac{3T^2}{2} & T \end{bmatrix} \quad (2)$$

where T denotes the radar revisit interval, and κ_q represents the process noise intensity of the q -th target.

B. Signal Model

Consider a C-MIMO radar system employing spatially colocated transmit and receive uniform linear array (ULA) with M antenna elements. Under the narrowband and far-field assumptions, the transmit and receive arrays share the same steering vector $\mathbf{a}(\theta) \in \mathbb{C}^{M \times 1}$.

At time k , the radar transmits L baseband snapshots within one dwell time, denoted by the waveform matrix $\mathbf{S}_k \in \mathbb{C}^{M \times L}$. The synthesized transmit signal vector at a spatial direction θ is given by [13], [15]:

$$\mathbf{x}_k(\theta) = \mathbf{a}^T(\theta) \mathbf{S}_k \quad (3)$$

Accordingly, the transmit beampattern at time k , defined as the spatial power distribution of the synthesized signal, is expressed as [1]:

$$P_k^{\text{Tx}}(\theta; \mathbf{R}_k) = \frac{1}{L} \mathbf{x}_k(\theta) \mathbf{x}_k^H(\theta) = \mathbf{a}^H(\theta) \mathbf{R}_k \mathbf{a}(\theta) \quad (4)$$

where $\mathbf{R}_k = \frac{1}{L} \mathbf{S}_k \mathbf{S}_k^H$ represents the waveform covariance matrix (WCM), satisfying the total energy constraint $\text{Tr}(\mathbf{R}_k) = E_{\text{total}}$.

Assuming that there are Q point targets in the surveillance region, the baseband echo signal received by the array at time k is modeled as:

$$\mathbf{y}_k(t) = \sum_{q=1}^Q \beta_q \mathbf{a}(\theta_{q,k}) \mathbf{a}^T(\theta_{q,k}) \mathbf{s}_k(t - \tau_{q,k}) e^{-j2\pi f_{q,k} t} + \mathbf{n}_k(t) \quad (5)$$

where β_q , $\theta_{q,k}$, $\tau_{q,k}$, and $f_{q,k}$ denote the complex reflection coefficient, azimuth angle, round-trip delay, and Doppler frequency of the q -th target, respectively. $\mathbf{n}_k(t)$ represents the zero-mean additive white Gaussian noise.

The receiver applies matched filtering and ideal adaptive beamforming. With receive gain fixed to $|\mathbf{w}^H \mathbf{a}(\theta)|^2 = M^2$, the detection energy depends solely on the transmit gain $P_k^{\text{Tx}}(\theta_{q,k}; \mathbf{R}_k)$. Consequently, $\mathbf{R}_k$ serves as the sole design variable.

C. Measurement Model

The observation vector $\mathbf{z}_{q,k}$ for the q -th target at time k is modeled as [5]:

$$\mathbf{z}_{q,k} = \mathbf{h}(\mathbf{x}_{q,k}) + \mathbf{v}_{q,k} \quad (6)$$

where $\mathbf{h}(\cdot)$ is the nonlinear measurement function. The measurement noise $\mathbf{v}_{q,k}$ is modeled as a zero-mean Gaussian vector with covariance matrix $\Sigma_{q,k}$, whose diagonal elements

represent error variances and are inversely proportional to the transmit beampattern gain:

$$\begin{cases} \sigma_{r,q}^2 \propto (\eta_q P_k^{\text{Tx}}(\theta_{q,k}; \mathbf{R}_k) \beta_{\text{bw}}^2)^{-1} \\ \sigma_{\theta,q}^2 \propto (\eta_q P_k^{\text{Tx}}(\theta_{q,k}; \mathbf{R}_k) \beta_{\text{ap}}^2)^{-1} \\ \sigma_{\tau,q}^2 \propto (\eta_q P_k^{\text{Tx}}(\theta_{q,k}; \mathbf{R}_k) \tau_{\text{eff}}^2)^{-1} \end{cases} \quad (7)$$

where β_{bw} , β_{ap} , and τ_{eff} denote the effective bandwidth, effective aperture length, and effective time duration of the transmit signal, respectively. The coefficient η_q aggregates the target RCS and path loss effects. Consequently, the measurement noise covariance matrix can be formulated in a compact explicit form:

$$\Sigma_{q,k}(\theta_{q,k}, \mathbf{R}_k) = [P_k^{\text{Tx}}(\theta_{q,k}; \mathbf{R}_k)]^{-1} \mathbf{D}_{q,k} \quad (8)$$

where $\mathbf{D}_{q,k}$ is a diagonal matrix containing waveform and environmental parameters, independent of $\mathbf{R}_k$. This indicates that increasing the transmit gain P_k^{Tx} at $\theta_{q,k}$ directly reduces the measurement error variances in $\Sigma_{q,k}$.

III. PROPOSED RTBD STRATEGY FOR MTT

A. Angular Uncertainty Modeling

To enable closed-loop tracking, the radar system continuously updates the target state estimates based on current measurements. Given the nonlinear measurement model described in Section II, we employ the EKF to perform recursive state estimation.

Utilizing the one-step prediction mechanism of the EKF, we obtain the predicted state vector $\hat{\mathbf{x}}_{q,k|k-1}$ and the corresponding prediction error covariance matrix $\mathbf{P}_{q,k|k-1}$ for the q -th target at time k . To quantify the spatial uncertainty, we extract the 2-D position sub-vector $\hat{\mathbf{x}}_{q,k|k-1}^{\text{pos}} \in \mathbb{R}^2$ and the associated 2×2 position covariance submatrix $\mathbf{P}_{q,k|k-1}^{\text{pos}}$. Under the Gaussian distribution assumption, the true target position $\mathbf{x}_{q,k}^{\text{pos}}$ resides within a confidence ellipsoid $\mathcal{E}_{q,k}$ with a probability level η [9], [16], [17]:

$$(\mathbf{x} - \hat{\mathbf{x}}_{q,k|k-1}^{\text{pos}})^T (\mathbf{P}_{q,k|k-1}^{\text{pos}})^{-1} (\mathbf{x} - \hat{\mathbf{x}}_{q,k|k-1}^{\text{pos}}) \leq \chi_{\eta,2}^2 \quad (9)$$

where $\chi_{\eta,2}^2$ denotes the threshold of the Chi-square distribution with 2 degrees of freedom. This ellipsoid characterizes the spatial uncertainty region of the target in the Cartesian coordinate system derived from the prior information.

According to the signal model in Section II, the effective echo energy is determined by the transmit beampattern gain $P_k^{\text{Tx}}(\theta; \mathbf{R}_k)$. Since the true target position is confined within the uncertainty ellipsoid $\mathcal{E}_{q,k}$, its azimuth angle $\theta_{q,k}$ is consequently uncertain. Geometrically, the range of feasible azimuth angles, denoted as $\Theta_{q,k}$, corresponds to the angular sector subtended by the two tangent lines from the radar origin to the ellipsoid $\mathcal{E}_{q,k}$. By solving for the geometric boundaries defined by these tangents, we determine the angular uncertainty interval:

$$\Theta_{q,k} = [\theta_{q,k}^{\min}, \theta_{q,k}^{\max}] \quad (10)$$

This interval $\Theta_{q,k}$ encapsulates the set of all possible azimuth angles for the target at the given confidence level. In the subsequent robust design, the true target azimuth $\theta_{q,k}$ is treated as an unknown variable belonging to $\Theta_{q,k}$, which directly impacts the actual transmit gain delivered to the target.

B. Derivation of Robust Bayesian Cramér-Rao Lower Bound

As a predictive performance metric, the Bayesian Cramér-Rao lower bound (BCRLB) is widely adopted for target tracking. Under the recursive BCRLB framework [9], [10], the predicted BCRLB matrix $\mathbf{C}_{q,k}^{\text{BCRLB}}$ for target q is approximated as the inverse of the Bayesian information matrix $\mathbf{J}_{q,k}$:

$$\mathbf{C}_{q,k}^{\text{BCRLB}}(\theta_{q,k}, \mathbf{R}_k) = [\mathbf{J}_{q,k}(\theta_{q,k}, \mathbf{R}_k)]^{-1} \quad (11)$$

where $\mathbf{J}_{q,k}$ can be expressed as:

$$\begin{aligned} \mathbf{J}_{q,k}(\theta_{q,k}, \mathbf{R}_k) = & \left[\mathbf{Q}_{k-1} + \mathbf{F}_{q,k-1} \mathbf{J}_{q,k-1}^{-1} \mathbf{F}_{q,k-1}^T \right]^{-1} \\ & + P_k^{\text{Tx}}(\theta_{q,k}; \mathbf{R}_k) \cdot \hat{\mathbf{H}}_{q,k}^T \mathbf{D}_{q,k}^{-1} \hat{\mathbf{H}}_{q,k} \end{aligned} \quad (12)$$

where $\hat{\mathbf{H}}_{q,k}$ is the Jacobian matrix of the nonlinear measurement function evaluated at the predicted state. For notational brevity, we define the prior information matrix as $\mathbf{J}_{q,k}^P = [\mathbf{Q}_{k-1} + \mathbf{F}_{q,k-1} \mathbf{J}_{q,k-1}^{-1} \mathbf{F}_{q,k-1}^T]^{-1}$ and the normalized data information matrix as $\mathbf{\Gamma}_{q,k} = \hat{\mathbf{H}}_{q,k}^T \mathbf{D}_{q,k}^{-1} \hat{\mathbf{H}}_{q,k}$.

However, since the true azimuth $\theta_{q,k}$ is uncertain within $\Theta_{q,k}$, the exact BCRLB cannot be computed directly. Based on the monotonic relationship in (12), the worst-case tracking performance exactly corresponds to the minimum transmit gain. Thus, we introduce γ_q to lower-bound the transmit gain $P_k^{\text{Tx}}(\theta; \mathbf{R}_k)$ across $\Theta_{q,k}$. Substituting this worst-case gain, the RBCRLB for the q -th target is defined as:

$$\tilde{\mathbf{C}}_{q,k}(\gamma_q) = \text{Tr} \left([\mathbf{J}_{q,k}^P + \gamma_q \cdot \mathbf{\Gamma}_{q,k}]^{-1} \right) \quad (13)$$

This equation establishes an analytical relationship between the worst-case tracking accuracy and the transmit gain lower bound γ_q , serving as a tractable convex performance metric for the subsequent robust beampattern design.

C. Formulation and Solution of the RTBD Problem

We adopt a Min-Max criterion to formulate the robust beampattern design problem. To make the continuous uncertainty interval $\Theta_{q,k}$ computationally tractable, we discretize it into a finite grid set $\tilde{\Theta}_{q,k} = \{\vartheta_{q,k}^{(i)}\}_{i=1}^{N_g}$. By minimizing the worst-case RBCRLB among all targets, this approach ensures tracking robustness and balances accuracy under angular uncertainty. Let the waveform correlation matrix $\mathbf{R}_k$ and the gain lower bounds $\{\gamma_q\}_{q=1}^Q$ be the optimization variables. The proposed RTBD optimization problem is formulated as:

$$\begin{aligned} \min_{\mathbf{R}_k, \{\gamma_q\}_{q=1}^Q} \quad & \max_{q \in \{1, \dots, Q\}} \tilde{\mathbf{C}}_{q,k}(\gamma_q) \\ \text{s.t.} \quad & \text{C1: } \mathbf{a}^H(\vartheta_{q,k}^{(i)}) \mathbf{R}_k \mathbf{a}(\vartheta_{q,k}^{(i)}) \geq \gamma_q, \quad \forall i, q, \\ & \text{C2: } \mathbf{R}_k \succeq \mathbf{0}, \\ & \text{C3: } \text{Tr}(\mathbf{R}_k) = E_{\text{total}}, \end{aligned} \quad (14)$$

where $i \in \{1, \dots, N_g\}$ and $q \in \{1, \dots, Q\}$.

The constraints are defined as follows:

- C1 ensures the transmit gain for target q at each grid point $\vartheta_{q,k}^{(i)}$ exceeds γ_q , thereby guaranteeing robustness against angular mismatches within the uncertainty set.
- C2 enforces the positive semi-definiteness of $\mathbf{R}_k$.
- C3 limits the total transmit power to E_{total} .

Convexity and Solution: Observing the structure of the objective function, the term $\tilde{\mathbf{C}}_{q,k}(\gamma_q) = \text{Tr}([\mathbf{J}_{q,k}^P + \gamma_q \mathbf{\Gamma}_{q,k}]^{-1})$ is the trace of the inverse of an affine matrix function. Since $\mathbf{J}_{q,k}^P$ and $\mathbf{\Gamma}_{q,k}$ are positive definite matrices, $\tilde{\mathbf{C}}_{q,k}(\gamma_q)$ is a convex and monotonically decreasing function with respect to γ_q . Furthermore, constraint C1 represents a set of linear inequalities with respect to $\mathbf{R}_k$ and γ_q , while C2 and C3 are standard convex constraints (linear matrix inequality and affine equality). Consequently, the optimization problem (14) is a convex Semidefinite Programming (SDP) problem. It can be efficiently solved using interior-point methods via standard convex optimization toolboxes such as CVX [18].

IV. SIMULATION RESULTS

In this section, we validate the effectiveness and robustness of the proposed RTBD strategy through numerical simulations. Consider a massive C-MIMO radar system equipped with a ULA consisting of $M = 100$ transmit/receive antennas with half-wavelength spacing. The total transmit energy is set to $E_{\text{total}} = M = 100$. The radar operates at a carrier frequency of $f_c = 10$ GHz with a tracking revisit interval of $T = 5$ s.

The surveillance scenario involves $Q = 3$ targets with a process noise intensity of $\kappa_q = 0.001$. The initial states are listed in Table I. To balance the beam robustness and peak gain, the confidence level is set to $\eta = 0.95$. The grid size is $N_g = 15$ to ensure spatial gain smoothness without redundant loss of degrees of freedom. Notably, the proposed RTBD exhibits more pronounced advantages over traditional algorithms as the target uncertainty jointly accumulated by the process noise κ_q and the revisit interval T increases. Simulations are conducted over $K = 10$ frames, with results averaged over $N_{\text{mc}} = 300$ Monte Carlo runs.

TABLE I
INITIAL STATES OF TARGETS

Target Index	x (km)	vx (km/s)	y (km)	vy (km/s)
Target 1	0	0	40	-0.6
Target 2	-10	-0.2	10	0.4
Target 3	10	0.2	10	0.4

Benchmark Algorithms: To evaluate the performance, the proposed RTBD strategy is compared with the following two baseline methods:

- Baseline 1: Non-Robust Transmit Beampattern Design (NR-TBD). This method treats the predicted angle as the deterministic true value, ignoring angular uncertainty. It optimizes the transmit covariance matrix to minimize the maximum BCRLB among all targets based on these point estimates.

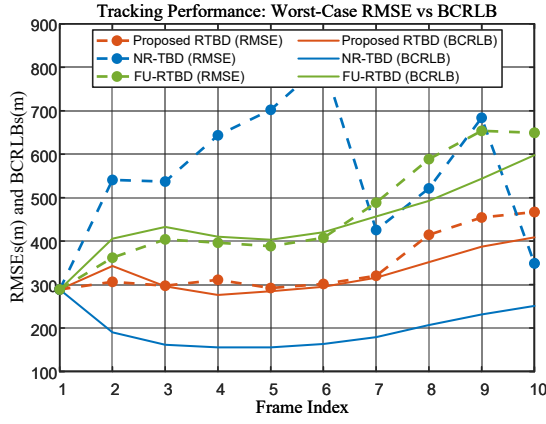


Fig. 1. Comparison of the worst-case target RMSE and BCRLB for different algorithms.

- Baseline 2: Fixed-Uncertainty Robust Transmit Beampattern Design (FU-RTBD). This method considers angular uncertainty but assumes a fixed uncertainty range of $\Delta\theta = \pm 4^\circ$ throughout the tracking process.

Regarding computational complexity, all three methods solve an SDP dominated by $\mathcal{O}(M^{6.5})$. Compared to NR-TBD, the proposed RTBD merely introduces a linear constraint overhead due to N_g . Therefore, its real-time execution performance is comparable to the baselines, remaining highly feasible for practical engineering applications.

Fig. 1 compares the BCRLB and RMSE of the worst-case target for the proposed RTBD, NR-TBD, and FU-RTBD strategies. Since the Min-Max criterion is adopted, we focus on the performance of the worst-case target. It can be observed that the Proposed RTBD strategy achieves the lowest RMSE curve. In contrast, although the NR-TBD method calculates the lowest theoretical BCRLB, its actual RMSE is relatively high and fails to align with its BCRLB curve. This severe mismatch arises because NR-TBD optimizes based solely on the angle of the predicted position. When the true target angle deviates from the predicted value, the actual transmit gain P_k^{Tx} significantly decays compared to the calculated value, causing a sharp increase in the RMSE. The FU-RTBD method maintains consistency between its BCRLB and RMSE through its robust design. However, due to the fixed, large uncertainty angular range, it wastes energy covering unnecessary spatial areas, resulting in both BCRLB and RMSE performance being inferior to the proposed RTBD strategy.

Fig. 2 illustrates the transmit beampatterns of the three algorithms at a specific frame. Evidently, the RTBD algorithm forms a mainlobe that robustly covers the true target angle. In contrast, the NR-TBD algorithm produces a narrow beam centered at the predicted angle; when prediction errors occur, the mainlobe may fail to align with the true target direction, potentially leading to severe gain loss for some targets. The FU-RTBD algorithm also covers the true target angle within its mainlobe, but its peak gain is noticeably lower than that of the RTBD algorithm due to unnecessary beam broadening.

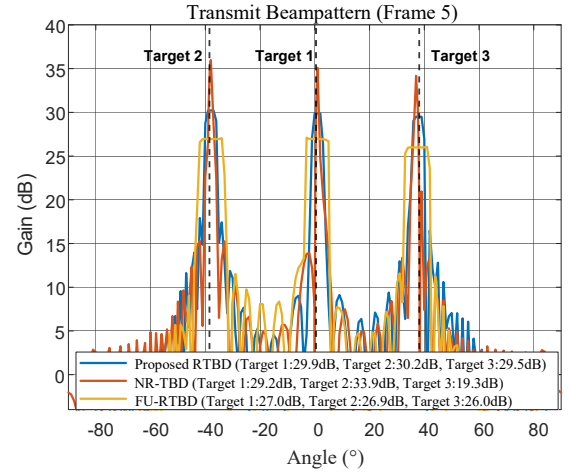


Fig. 2. Comparison of transmit beampatterns at a specific frame.

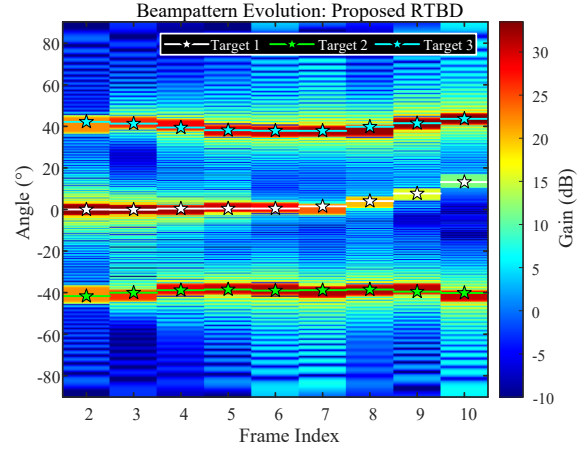


Fig. 3. Transmit beampattern evolution of the proposed RTBD algorithm.

Fig. 3 displays the transmit gain of the proposed RTBD algorithm at the true target angles across different frames. It can be seen that throughout all tracking frames, the targets are consistently located within the high-gain mainlobe region. This demonstrates that the proposed strategy possesses excellent robustness, effectively accommodating the angular uncertainty during the dynamic tracking process.

V. CONCLUSION

In this paper, an RTBD strategy is proposed for MTT in C-MIMO radar under angular uncertainty. By leveraging the prediction error covariance provided by the Kalman filter, adaptive angular uncertainty sets are constructed for each target, and the RBCRLB is derived as the tracking performance metric. By discretizing the continuous uncertainty intervals, the RTBD optimization problem is formulated as a convex SDP, which can be solved efficiently. Simulation results demonstrate that the RTBD method can adaptively adjust its beamwidth in response to angular uncertainty, effectively mitigating the transmit gain degradation caused by target estimation errors.

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